



SYSTEM REQUIREMENT SPECIFICATION (SRS)

P.R.I.S.M

Predictive Real-time Intelligence Sentiment Monitoring System

Software-as-a-Service Platform

DOCUMENT CONTROL

FIELD	DESCRIPTION
Document Title	PRISM_SRS_SaaS_v1.0
Version	1.0
Prepared By	Aras Integrasi
Customer	Bahagian Keselamatan Strategik - Jabatan Perdana Menteri
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Approval Authority	Farul Mohd Ghazali (Aras Integrasi)

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1. INTRODUCTION

1.1 Purpose

This System Requirement Specification (SRS) defines the system-level functional and non-functional requirements of PRISM, delivered by Aras Integrasi as a subscription SaaS. It translates the user requirements in PRISM_URS_SaaS_v1.0 into verifiable system behaviour, and is intended for the Customer's technical reviewers and the Aras engineering team. Every system requirement is tagged with the platform role(s) it serves (L = Leader, P = PMO/Analyst, S = SME, A = System Administrator) and traced to its parent user requirement(s).

1.2 Scope

The system comprises a browser-based War Room application (Command, Intelligence, Operations and System Administration layers), an AI agent fleet on an orchestration runtime, a multi-store data platform (relational, graph and vector), ingestion pipelines, a prediction and simulation engine, a conversational AI service with retrieval-augmented generation, and a reporting engine. The platform is multi-deployment by design: each Customer engagement is an isolated, configuration-driven tenant. No election-, region- or client-specific logic is hard-coded.

1.3 Definitions

TERM	DEFINITION
Area of Interest	A configured geographic or administrative unit (e.g. constituency, district, region) against which scoring, alerting and reporting operate.
Watched Entity	A configured person or organisation (e.g. candidate, public figure) tracked continuously for sentiment, mentions and issues.
Agent	An autonomous scheduled worker on the orchestration runtime executing a defined pipeline (ingestion, scoring, alerting, etc.).
Heartbeat	The scheduled interval at which an agent's workflow executes.
Data Lake	The governed store holding raw history and refined layers; single source of truth serving all screens.
RAG	Retrieval-Augmented Generation — the conversational AI retrieves evidence from the vector store before generating an answer.
Sensitive Category	Content classes designated by the Customer as high-risk (e.g. 3R-type issues) subject to human-only handling.

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2. SYSTEM ARCHITECTURE

2.1 Logical Layers

LAYER	COMPONENTS
Command Layer	Executive Dashboard; Command Chat (conversational AI); Geospatial Heatmap with per-area detail; Narrative Monitor (news sentiment, social pulse, topic network); Scenario Simulation.
Intelligence Layer	Area Watch; Entity Profiler; Entity Watchlist; Segment Analysis; Ground-Intelligence Results.
Operations Layer	Knowledgebase; Resource Optimizer; Field-Team Manager; Agent Management; SME Predictions / ground input.
System Admin Layer	User Management (RBAC); Data Connections & Register; Ingestion (Scrape) Schedules; Audit Logs.
Field Layer (v2.0)	Hard-isolated field portal for canvassing submission (separate authentication and UI; shared backend), deferred to v2.0.

2.2 Data Platform

The platform operates a collect-once, refine-in-stages, serve-everywhere data cascade: (1) COLLECT — news feeds, social platforms (where subscribed), ground input and static registers arrive on their own schedules, some at 5-minute intervals; (2) REFINE — deduplication, translation, sentiment scoring, issue tagging and area linking; (3) LAKE — one governed store holding raw history and refined layers, preserving point-in-time history; (4) SERVE — fast serving layers feed every dashboard, the conversational AI and the prediction engine.

STORE	ROLE IN THE PLATFORM
Relational store (PostgreSQL)	Primary structured store: articles, sentiment snapshots, ground sessions, predictions, accounts, agent run logs, schedules, segment scores.
Graph store (Neo4j)	Topic, area and actor nodes with co-occurrence and relationship edges; powers the Topic Network and influence mapping.
Vector store (Qdrant)	Embeddings of articles, knowledgebase documents and ground notes; powers semantic search and RAG citations in Command Chat.

2.3 AI & ML Components

COMPONENT	FUNCTION
Prediction model (anchored ensemble + LLM reasoning)	Per-area outcome projection anchored to configured historical baselines, adjusted by live sentiment, demographics and ground returns; LLM component performs voter/audience behaviour simulation.
MTizer sentiment engine	Proprietary Malay/English sentiment scoring trained on local political and public-affairs content; per-item scores with confidence.
MEMber Suara STT	Malay-primary speech-to-text for voice-based ground reporting.
MiroFish content integrity	Fact-checking and misinformation/inauthentic-behaviour verification supporting the Content Integrity agent.
Eternis runtime	Agent orchestration: scheduling (heartbeats), pipeline execution, inter-agent events, run logging.

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3. FUNCTIONAL SYSTEM REQUIREMENTS

Requirement conventions: each requirement is verifiable, role-tagged, and traced to parent user requirement(s) in PRISM_URS_SaaS_v1.0.

3.1 Ingestion & Data Pipeline (SR-ING)

ID	SYSTEM REQUIREMENT	ROLE(S)	TRACES TO	VERIFICATION
SR-ING-01	The system shall poll configured news feeds at configurable intervals (default 5 minutes), filter by configured keyword sets, and deduplicate against previously ingested items before persistence.	A	UR-ADM-03	Run logs; duplicate-skip test
SR-ING-02	The system shall score every ingested item for sentiment (-1 to +1) with per-item confidence, in Malay and English, and persist score snapshots with provenance.	P	NFR-03, NFR-04	Sample scoring audit
SR-ING-03	The system shall tag every ingested item into a hierarchical, configurable issue taxonomy and link items to areas of interest and watched entities.	P	UR-INT-04/07	Taxonomy tagging test
SR-ING-04	The system shall ingest social-platform data through pluggable provider connectors when subscription credentials are configured, with per-platform breakdowns.	P, A	UR-CMD-05, C-2	Connector activation test
SR-ING-05	The system shall accept structured ground input (SME submissions, complaint/programme records with sub-categories) and make it available to analysis the same day.	S, P	UR-SME-03/05	End-to-end submission test
SR-ING-06	The system shall detect keyword/topic volume spikes ahead of mainstream coverage and propose new keywords for the ingestion schedules.	P, A	UR-INT-06	Spike-detection replay
SR-ING-07	The system shall preserve raw ingestion history so that point-in-time questions ("what did we know on date X?") are answerable per the retention policy.	P, A	NFR-10	Point-in-time query test

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3.2 AI Agent Fleet (SR-AGT)

ID	SYSTEM REQUIREMENT	ROLE(S)	TRACES TO	VERIFICATION
SR-AGT-01	The system shall operate all autonomous workers on an orchestration runtime providing scheduling, pipeline execution, inter-agent events and run logging with per-agent success metrics.	A	UR-ADM-05	Console walkthrough
SR-AGT-02	The fleet shall include, at minimum: news curation, prediction, escalation, content integrity, messaging simulation, topic tagging and trend detection agents, each independently start/stop-able and configurable.	A	UR-ADM-05	Per-agent control test
SR-AGT-03	The escalation agent shall evaluate projections and ground data against configurable thresholds and generate structured alerts (severity, category, recommended action, time-sensitivity).	L, P	UR-CMD-08	Threshold-breach test
SR-AGT-04	The content-integrity agent shall flag misinformation and coordinated inauthentic behaviour with severity and suggested response, and shall be able to trigger escalation.	P	UR-CMD-08	Flagging test with seeded content
SR-AGT-05	Sensitive-category content shall route exclusively to human escalation paths; the system shall provide no automated public response mechanism for such content.	All	UR-CMD-08, NFR-05	Code/config inspection; negative test
SR-AGT-06	An administrative kill-switch shall suspend any agent, output stream or distribution immediately, and the action shall be audit-logged.	A, L	UR-ADM-06	Controlled kill-switch drill

3.3 Prediction & Simulation Engine (SR-PRD)

ID	SYSTEM REQUIREMENT	ROLE(S)	TRACES TO	VERIFICATION
SR-PRD-01	The engine shall score every configured area on a schedule (default hourly) and	L, P	UR-INT-01	Refresh-cycle verification

ID	SYSTEM REQUIREMENT	ROLE(S)	TRACES TO	VERIFICATION
	event-driven upon significant sentiment shifts, so briefings never rely on stale numbers.			
SR-PRD-02	Projections shall merge configurable inputs: historical baselines (multiple prior cycles), demographics, live sentiment, entity-acceptance and organisational-strength factors, and validated ground returns.	P	UR-INT-08	Model-input audit
SR-PRD-03	The composite index shall support time-decaying weights per input class (e.g. baseline ~40% early-cycle decaying to ~30%) and an explicit uncertainty (omega) adjustment in the range -10 to +10 with recorded rationale.	P	UR-INT-08	Config + output inspection
SR-PRD-04	The simulation service shall support 1-way, 2-way and 3-way contest configurations with per-area starting-environment overrides, issue-weight sliders, turnout adjustments and sentiment overrides.	L, P	UR-CMD-06	Scenario matrix test
SR-PRD-05	Simulations shall execute in a sandbox and shall never write to live projection data; simulation outputs shall be visibly labelled as simulated.	All	C-5	Isolation test
SR-PRD-06	The messaging-simulation service shall model segment-level reactions to a proposed message across the four configured strategy modes, and require recorded human approval before results are actioned.	L, P	UR-CMD-07	Mode-by-mode run; approval gate test

3.4 Conversational AI & Knowledge Services (SR-CHT)

ID	SYSTEM REQUIREMENT	ROLE(S)	TRACES TO	VERIFICATION
SR-CHT-01	Command Chat shall answer natural-language questions by retrieving evidence from the vector store and structured stores (RAG) and shall display	L, P	UR-CMD-03, NFR-04	Citation spot-checks

ID	SYSTEM REQUIREMENT	ROLE(S)	TRACES TO	VERIFICATION
	citations: sources, counts and refresh time.			
SR-CHT-02	The system shall auto-generate a Daily Brief on session start covering sentiment movement, risk changes, ground highlights and recommended actions, without user prompting.	L, P	UR-CMD-03	Session-start observation
SR-CHT-03	Knowledgebase uploads shall be auto-tagged by area and entity, embedded into the vector store, and citable in answers as SME insight.	S, P	UR-SME-01	Upload-to-citation test
SR-CHT-04	Users with the SME role shall be able to convert a chat answer or search result into a persistent, attributed source note that joins the knowledgebase corpus.	S	UR-SME-02	Note-creation test
SR-CHT-05	The guided-interview service shall walk an analyst through a configurable analytical framework (e.g. R.I.S.I.K), pulling platform evidence per step and emitting a structured intelligence report with a conclusion.	P, S	UR-INT-09	End-to-end guided run

3.5 Visualisation & Dashboards (SR-VIS)

ID	SYSTEM REQUIREMENT	ROLE(S)	TRACES TO	VERIFICATION
SR-VIS-01	The Executive Dashboard shall render the widget set defined in UR-CMD-01 from the serving layer without manual refresh, with role-configurable widget order and default landing pages.	L, P, A	UR-CMD-01/02	Dashboard walkthrough
SR-VIS-02	The geospatial heatmap shall colour-code every configured area by projection band, refresh colours at least daily, and drill into per-area detail (projection, sentiment history, baseline, demographics, sub-district table, recent items).	L, P	UR-CMD-04	Map + detail walkthrough
SR-VIS-03	The system shall provide an adjacency view showing neighbouring areas' mutual	P	UR-CMD-04	Adjacency render test

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ID	SYSTEM REQUIREMENT	ROLE(S)	TRACES TO	VERIFICATION
	influence for a selected area, supporting spatial pattern analysis by analysts.			
SR-VIS-04	The topic network view shall render topic/area/actor relationships from the graph store, with a narrative-gaps panel identifying discussed-but-unaddressed topics.	P	UR-INT-07	Graph walkthrough
SR-VIS-05	All lists and tables central to operations (areas, entities, sessions, alerts) shall be filterable and sortable, with per-area and per-window scoping.	L, P	UR-INT-01/03/05	Filter/sort test

3.6 Reporting Engine (SR-RPT)

ID	SYSTEM REQUIREMENT	ROLE(S)	TRACES TO	VERIFICATION
SR-RPT-01	The reporting engine shall generate scheduled reports per the configured cadence (default: daily ~09:00 and ~17:00, weekly narrative analysis, periodic deep dives) with configurable recipients per report.	L, P, A	UR-RPT-01	Schedule + delivery test
SR-RPT-02	Scheduled reports shall render to distribution-quality PDF and deliver to configured channels/groups automatically.	L, P, A	UR-RPT-02	PDF render + delivery test
SR-RPT-03	Every generated report, dossier and exported artefact shall automatically carry the standard disclaimer limiting liability and stating the advisory nature of AI outputs.	All	UR-RPT-03	Artefact inspection
SR-RPT-04	Report content shall include, at minimum: overnight issues and incidents, trend/pattern monitoring, risk and threat detection, narrative suggestion and avoidance lists, and recommended actions.	L, P	UR-RPT-04	Content checklist review

3.7 Administration, Security & Governance (SR-ADM)

ID	SYSTEM REQUIREMENT	ROLE(S)	TRACES TO	VERIFICATION
SR-AD M-01	The system shall enforce role-based access control across every module and API; role scopes shall be centrally administered; deactivation shall take effect immediately.	A	UR-ADM-01, NFR-02	RBAC matrix test
SR-AD M-02	Field/SME-facing surfaces shall be architecturally separated from leadership modules (separate authentication context), such that cross-access is impossible rather than merely hidden.	A	NFR-02	Cross-access negative test
SR-AD M-03	The in-platform data register shall record, for every dataset/connection: owner, freshness class (live / weekly / per-cycle / snapshot), validation status and live health (connected / pending / stale).	A, P	UR-ADM-02	Register walkthrough
SR-AD M-04	All significant events (authentication, configuration changes, exports, agent runs, approvals, kill-switch use) shall be written to a tamper-evident (hash-chained) audit log retained per policy.	A	UR-ADM-04, NFR-08	Hash-chain verification
SR-AD M-05	Designated critical actions shall require two-person approval enforced by the system; the approval pair shall be recorded in the audit log.	L, A	UR-ADM-07	Dual-approval test
SR-AD M-06	Credentials and keys shall reside only in secured configuration stores; the system shall expose no secrets in UI, logs, exports or documentation.	A	NFR-07	Secret-scan audit
SR-AD M-07	Personal data shall be processed per PDPA 2010: aggregation/anonymisation by design for individual-level records; extra governance flags for sensitive categories (welfare, finances, faith) before entering any analysis.	A	NFR-07	Data-protection review
SR-AD M-08	Per-dataset-class retention and post-cycle archival shall be configurable and executable as a documented archival job.	A	NFR-10	Archival dry-run

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4. EXTERNAL INTERFACES

INTERFACE	DIRECTION / PROTOCOL	ACTIVATION CONDITION
News feeds (RSS/API)	Inbound; polled per schedule	Active by default with configured source list
Social-listening provider	Inbound; provider connector	Customer-supplied subscription credentials
Official electoral / registry data	Inbound; provider API	Customer-obtained credentials and lawful basis
Resource-management backend	Inbound; API	Customer system availability; enables live resource optimisation and financial tracking
Report distribution channels	Outbound; PDF to configured channels/groups	Configured recipients; content-moderation policy for any public-facing distribution
Speech-to-text service	Internal; audio upload or live capture	Active for voice-based ground input

5. NON-FUNCTIONAL SYSTEM REQUIREMENTS

ID	REQUIREMENT	TARGET / VERIFICATION
SNF-01	Performance: interactive pages load within ~2–3 s under normal load; dashboards auto-refresh from the serving layer.	Timed walkthrough at reference load
SNF-02	Scale: sustained ingestion and scoring of thousands of items/day; hundreds of watched entities; all configured areas scored hourly without degradation.	Load test at reference volumes
SNF-03	Availability: 24/7 operation with managed monitoring, backups and recovery as part of the SaaS service.	Service-level review; backup restore drill
SNF-04	Multi-tenancy & isolation: each Customer engagement operates as an isolated tenant; no data crosses tenants.	Tenant-isolation test
SNF-05	Configurability: areas, entities, taxonomies, thresholds, cadences, weights and languages configurable without code change; onboarding in weeks.	Onboarding runbook execution
SNF-06	Localisation: Malay and English processed natively across text and voice, including local idiom; per-item confidence recorded.	Bilingual scoring sample
SNF-07	Explainability: every AI output traceable to sources, model and refresh time (provenance).	Provenance spot-checks
SNF-08	Human-in-the-loop: no autonomous execution of spend, deployment or public messaging anywhere in the system.	Architecture review; negative tests
SNF-09	Security: encrypted transport; secured configuration for secrets; hardened managed infrastructure; access logging.	Security review
SNF-10	Maintainability: agents, connectors and report templates independently deployable; feedback-cycle releases without service interruption.	Release-process review

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6. CONSTRAINTS & DEPENDENCIES

REF	CONSTRAINT / DEPENDENCY
D-1	Third-party data feeds (official electoral/registry data, social listening, resource backends) activate only with Customer-supplied subscriptions, credentials and lawful basis.
D-2	Prediction quality depends on the configured historical baselines and the freshness of demographic and ground data supplied or validated by the Customer (SME validation workflow).
D-3	Automated message distribution remains disabled until the Customer approves a content-moderation workflow; the Message agent stays stopped until then.
D-4	Single sign-on is not part of the current build; authentication is username/password pending the SSO roadmap item.
D-5	The service operates on Aras-managed shared infrastructure unless a dedicated environment is subscribed.
D-6	News-source terms of use must permit ingestion for the subscription period.

7. TRACEABILITY MATRIX (UR → SR)

USER REQUIREMENT	SATISFIED BY SYSTEM REQUIREMENT(S)
UR-CMD-01 / 02	SR-VIS-01
UR-CMD-03	SR-CHT-01, SR-CHT-02
UR-CMD-04	SR-VIS-02, SR-VIS-03
UR-CMD-05	SR-ING-04, SR-VIS-04
UR-CMD-06	SR-PRD-04, SR-PRD-05
UR-CMD-07	SR-PRD-06
UR-CMD-08	SR-AGT-03, SR-AGT-04, SR-AGT-05
UR-INT-01 / 03 / 05	SR-PRD-01, SR-VIS-05
UR-INT-04 / 07	SR-ING-03, SR-VIS-04
UR-INT-06	SR-ING-06
UR-INT-08	SR-PRD-02, SR-PRD-03
UR-INT-09	SR-CHT-05
UR-SME-01 / 02	SR-CHT-03, SR-CHT-04
UR-SME-03 / 05	SR-ING-05
UR-SME-04	SR-PRD-02 (validated inputs), SR-ADM-03
UR-RPT-01 – 04	SR-RPT-01 – SR-RPT-04
UR-ADM-01 – 07	SR-ADM-01 – SR-ADM-08, SR-AGT-01/02/06
NFR-01 – NFR-10	SNF-01 – SNF-10

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8. SIGN-OFF

For and on behalf of Aras Integrasi Sdn. Bhd.	For and on behalf of the Bahagian Keselamatan Strategik, Jabatan Perdana Menteri
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Name:

Title:

Date:

Name:

Title:

Date: